

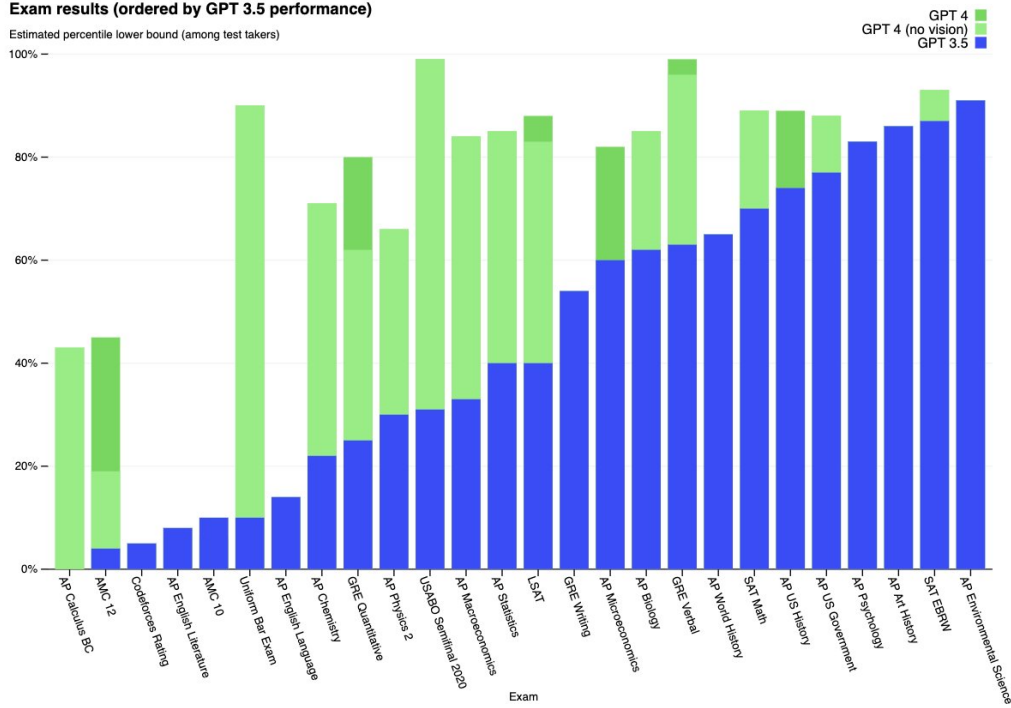


Figure 1: Taken directly from GPT-4 Technical Report (OpenAI, 2023b). To get a sense of how quickly model capabilities are progressing – consider the jump in exam performance between GPT-3.5 and GPT-4 (OpenAI, 2023b).

Our study is motivated less by the progress of these models alone though, and more by the breadth, scale, and capabilities we’ve seen in the complementary technologies developed around them. The role of complementary technologies remains to be seen, but maximizing the impact of LLMs appears contingent on integrating them with larger systems (Bresnahan, 2019; Agrawal et al., 2021). While the focus of our discussion is primarily on the generative capabilities of LLMs, it is important to note that these models can also be utilized for various tasks beyond text generation. For example, embeddings from LLMs can be used for custom search applications, and LLMs can perform tasks such as summarization and classification where the context may be largely contained in the prompt.

To complement predictions of technology’s impacts on work and provide a framework for understanding the evolving landscape of language models and their associated technologies, we propose a new rubric for assessing LLM capabilities and their potential effects on jobs. This rubric (A.1) measures the overall exposure of tasks to LLMs, following the spirit of prior work on quantifying exposure to machine learning (Brynjolfsson et al., 2018; Felten et al., 2018; Webb, 2020). We define exposure as a proxy for potential economic impact without distinguishing between labor-augmenting or labor-displacing effects. We employ human annotators and GPT-4 itself as a classifier to apply this rubric to occupational data in the U.S. economy, primarily sourced from the O*NET database.¹²

To construct our primary exposure dataset, we collected both human annotations and GPT-4 classifications, using a prompt tuned for agreement with a sample of labels from the authors. We observe similar agreement

¹This is distinct from recent social science research that makes use of LLMs to simulate human behavior (Horton, 2023; Sorensen et al., 2022)

²While our exposure rubric does not necessarily tie the concept of language models to any particular model, we were strongly motivated by our observed capabilities of GPT-4 and the suite of capabilities we saw in development with OpenAI’s launch partners (OpenAI, 2023b).

or task while maintaining consistent quality. We anticipate that adoption will be highest and most immediate for applications that realize a considerable increase in productivity. Although this threshold is somewhat arbitrary, it was selected for ease of interpretation by annotators. Moreover, regardless of the chosen threshold, we guessed that the real-world reduction in task time would likely be slightly or significantly lower than our estimates, leading us to opt for a relatively high threshold. In our own validation labeling, we found that this corresponded closely to whether an LLM or LLM-powered software could perform the core part of a task or nearly the entire task.

Comparison	γ	Weighting	Agreement	Pearson's
GPT-4, Rubric 1; Human	α	E1	80.8%	0.223
	β	E1 + .5*E2	65.6%	0.591
	ζ	E1 + E2	82.1%	0.654
GPT-4, Rubric 2; Human	α	E1	81.8%	0.221
	β	E1 + .5*E2	65.6%	0.538
	ζ	E1 + E2	79.5%	0.589
GPT-4, Rubric 1; GPT-4, Rubric 2	α	E1	91.1%	0.611
	β	E1 + .5*E2	76.0%	0.705
	ζ	E1 + E2	82.4%	0.680

Table 2: Model and human comparison of agreement and Pearson's correlation scores. The agreement score is determined by looking at how often the two groups agree on the annotation (e.g. E0, E1 or E2). In the paper we use GPT-4, Rubric 1. Core tasks are given twice the weight at the occupation-level as supplemental tasks. All weights sum to one.

We then collected both human and GPT-4-generated annotations using the exposure rubric, which underlie the bulk of the analyses in this paper.

- *Human Ratings:* We obtained human annotations by applying the rubric to each O*NET Detailed Worker Activity (DWA) and a subset of all O*NET tasks and then aggregated those DWA and task scores⁵ at the task and occupation levels. The authors personally labeled a large sample of tasks and DWAs and enlisted experienced human annotators who have reviewed GPT-3, GPT-3.5 and GPT-4 outputs as part of OpenAI's alignment work (Ouyang et al., 2022).
- *GPT-4 Ratings:* We administered a similar rubric to an early version of GPT-4 (OpenAI, 2023b) but on all task/occupation pairs rather than DWAs. We made slight modifications to the rubric (which was used as a "prompt" to the model in this case) to enhance agreement with a set of human labels. Full agreement rates are given in Table 2.

We construct three primary measures for our dependent variable of interest: (i) α , corresponding to E1 in the exposure rubric above, anticipated to represent the lower bound of the proportion of exposed tasks within an occupation, (ii) β , which is the sum of E1 and 0.5*E2, where the 0.5 weight on E2 is intended to account for exposure when deploying the technology via complementary tools and applications necessitates additional investment, and (iii) ζ , the sum of E1 and E2, an upper bound of exposure that provides an assessment of maximal exposure to an LLM and LLM-powered software. We summarize agreement between annotation groups and measures in Table 2. For the remainder of the analysis, if not specified, the reader may assume that

⁵The authors annotated DWAs that clearly required a high degree of physicality or manual dexterity, and the contracted annotators labeled the remaining activities, along with a subset of tasks including those without associated DWAs and those for which there was no clear task-level annotation after aggregating the DWA annotations.